

SEMESTER 2

MNU

Student guide



2025-2026

Faculty of medicine

Dear Semester 2 Medical School students,

We are so glad to welcome you to the new semester at Mansoura National University.

You are about to begin one of the most exciting times in your medical life at MNU.

By starting semester 2, you are about to gain a profound knowledge about our human body in a rapidly evolving field. We promise that your time will be cheerful as we are trying hard to make your journey a joyful experience.

The faculty members and the team are excited to welcome you. We are dedicated to making your learning experience a non-forgettable one.

Take a deep breath. No matter how long the years of studying it will pay off at the end.

Effort, discipline, and commitment are the keys to your success, and we promise that no effort will be saved to help you achieve your goals.

Wish you all a wonderful semester and always remember, education is an amazing opportunity for a great future.

Aya Elnegery

Semester 2 Chief Tutor

Amira Taman

**Director of Medicine and Surgery
Program, MNU**



Semester 2 composition

Semester 2 has five modules (PPPM, MSS, HIS, ECE 1 and UNI).

Course	Exam Marks						Total
	Credit point	Mid-term	Incision	Activity & portfolio & PBL & training	OSCE	Final	
Principles of pathology, microbiology, parasitology and pharmacology (PPPM - 201)	13	47	3	9	58	78 (45MCQ,33SAQ)	195
Musculo-skeletal system (MSS -202)	9	20	12	9	40	54 (25MCQ, 29SAQ)	135
Hematopoietic and Immune system (HIS -203)	4	10	5	3	18	24 (15MCQ,9SAQ)	60
Early Clinical Experience 1 (ECE 1 -204)	2	5		4	9	12 (8MCQ,9SAQ)	30
Contemporary Issues (UNI - 205)	2					30	30 P/F
Total	30						420

Program of Medicine and Surgery

Agenda for Semester 2- Level 1, 2025/2026

Week	Date	Module / Subject				Notes
1	07-12/02/2026	Musculoskeletal system	Hematopoietic and immune system	PPPM	Early clinical experience 1	
2	14-19/02/2026					Ramadan 18/02/2026
3	21-26/02/2026					
4	28/02 -05/03/2026					
5	07-12/03/2026					
	8/3/2026	Musculoskeletal system (MSS) mid Exam				
6	14-19/03/2026					
	15/3/2026	Hematopoietic and immune system (HIS) mid Exam				
	21/03-22/03/2026					Eid El Fetr
7	23-26/03/2026					
	24/3/2026	Principles of pathology, microbiology, parasitology and pharmacology (PPPM) mid Exam				
8	28/03-02/04/2026					
	29/3/2026	Early clinical experience 1(ECE 1) mid Exam				
9	04-09/04/2026	MSS	HIS	PPPM	ECE 1	
10	11-16/04/2026					13/04 Easter
11	18-23/04/2026					20-21/04 Faculty conference
12	25-30/04/2026					25/04 Sinai liberation Day & 1/05 Labor Day
13	02-07/05/2026					
14	09-14/05/2026					
15	16-21/05/2026					
16	23-28/05/2026					26-29/05 Eid Adha
17-22	30/05-09/07/2026	Final Exams				17/06 Hijri New Year 30/06 Revolution

Kindly pay attention that **Exam dates are provisional.**

Goal of Semester 2

The goal of semester 2 is to provide students with a strong foundation of basic sciences. Systems-based modules are centred on the various organ systems of the body and focuses on normal and abnormal structure and function. Anatomy, histology, physiology, biochemistry, microbiology, pharmacology, pathology, and parasitology are taught in an integrated fashion. Important clinical conditions related to each system are also introduced, so that the students can correlate clinical presentation with the pathophysiology. There will be weekly online quizzes starting from week 2. The quiz will be posted every Thursday.

Anatomy & Embryology

Description:

Anatomy is a part of basic science, where the student will learn the structure of human body. Anatomy consists of lectures followed by practical small-group sessions throughout 9 weeks to discover the magic of human anatomy. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Embryology is a part of science dealing with the development of the embryo. Embryology is challenging for many students because the body's organization as it develops is very often completely different from the final product with which we are all much more familiar. Embryology will be taught as small groups, so the students will have the opportunity to ask questions and doing activities. Embryology exam will be included with the anatomy exam.

Topics:

	Week No	Lectures	Practical
MSS module	1	Muscles of the back & arm muscles	Muscles of the back & arm muscles
	2	Shoulder girdle, shoulder joint & axilla	
	3	Anterior and posterior forearm muscles	Elbow, cubital fossa and wrist Muscles of the hand
	4	Nerves of the upper limb & injuries	Vessels of the upper Limb
	5	Muscles of the gluteal region and posterior thigh	
	6	Muscles of the anterior and medial thigh	
	7	Muscles of the leg and foot	
	8	Development and anomalies of vertebral column & limbs	
	9	Nerves of the lower limb	Spaces & vessels of the lower limb

Histology

Description:

Histology or microanatomy is a part of basic science, which studies the cell and its detailed structures. Histology will be introduced as lectures, practical and discussion small-group sessions throughout 9 weeks. Students will use the microscope to identify structures of the cell and human body. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
HIS Module	1	Structure of blood cells	
	2	Haemopoiesis & bone marrow	Blood (preparation of blood film)
	3	Lymphatic system 1	Blood (preparation of blood cells)
	4	Lymphatic system 2	lymphatic
MSS module	5	Histology of cartilage	The histological features of cartilage.
	6	Histology of bone 1	
	7	Histology of bone 2	The histological features of bone.
	8	Histology of muscle 1	
	9	Histology of muscle 2	The histological features of muscles.

Physiology

Description:

It is a basic science studying the normal function of the human body. Physiology will be introduced as lectures and practical small-group sessions throughout 11 weeks. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
HIS Module	1	Plasma proteins	Blood groups
	2	RBCS, count, structure, functions, sites of erythropoiesis	Hb, Hct, blood indices
	3	Functions of cellular elements of blood and factors affecting erythropoiesis.	Hemostasis and ESR
	4	Physiological basis of hemostatic mechanisms.	
	5	Natural anticoagulants and fibrinolysis.	
MSS module	6	Neuromuscular Transmission	
	7	Myasthenia gravis and functional histology of skeletal muscles.	Simple muscle twitch 1
	8	Mechanism of skeletal muscle contraction.	Simple muscle twitch 2
	9	Metabolic aspect of muscle exercise.	
	10	Types of muscle contraction and muscle fibers.	
	11	Smooth muscles.	

Biochemistry

Description:

Description:

Biochemistry is a part of basic science, exploring the chemical processes occurring within the human body and studying the structure of carbohydrates, lipids, proteins and nucleotides and, in turn, their functions and ways to control them. Biochemistry also describes the common biochemical tests used in diagnosis and treatment of human diseases. Biochemistry will be introduced as lectures and practical small-group laboratory sessions throughout 3 weeks. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
HIS Module	1	Haem synthesis	Assessment bilirubin of serum
	2	Hemoglobin structure and destruction	Pipetting
	3	Plasma proteins	

Pharmacology

Description:

Pharmacology is a basic science studying the drug properties, mode of action and drug interaction. Pharmacology will be learned as lectures and practical small-group sessions throughout 12 weeks. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
PPPM Module	1	Pharmacokinetics I	
	2	Pharmacokinetics II	Sources, dosage forms and Routes of administration
	3	Pharmacodynamics I Pharmacodynamics II	Prescription
	4	Parasympathomimetic drugs	Adverse drug reactions (ADRs)
	5	Parasympatholytic drugs	Applied General Pharmacology
	6	Sympathomimetic drugs Sympatholytic drugs	Eye pharmacology
HIS Module	7	Anticoagulants	
	8	Fibrinolytics, antiplatelets and drugs used in bleeding disorders	
	9	Drugs for allergy and autoimmunity	Applied Blood Pharmacology
MSS module	10	Analgesics and Antipyretics	
	11	Disease-modifying anti-rheumatic drugs (DMARDs)	Applied Pharmacology of Analgesics
	12	Skeletal muscle relaxants	Pharmacotherapy of Gout

Pathology

Description:

Pathology is the study of the causes and effects of the disease or injury. Pathology will be introduced as lectures, practical and discussion small-group sessions thought 13 weeks. Students will use the microscope to identify structures of the cell and human body. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
PPPM Module	1	Introduction to pathology Adaptation and reversible cell injury	Adaptation and reversible cell injury
	2	Irreversible cell injury	Irreversible cell injury
	3	Tissue accumulation	Tissue accumulation
	4	Acute inflammation Chronic inflammation	Inflammation
	5	Repair	Repair
	6	Introduction to Neoplasia	Benign tumors
	7	Etiology of Cancer	Malignant tumor
HIS Module	8	Pathology of lymphatic system	Lymph nodal and spleen disorders
MSS module	9	Pathology of joint diseases	
	10	Osteodystrophy	
	11	Osteomyelitis	
	12	Bone tumors	Pathology of bone diseases and bone tumors
	13	Soft tissue tumors	Soft tissue masses and tumors

Microbiology

Description:

Microbiology is the study of microorganisms, such as bacteria, viruses, archaea, fungi and protozoa and their effects on other living organisms. Microbiology will be learned as lectures and practical small-group sessions throughout 9 weeks. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
PPPM Module	1	Cells of the immune system	Sample taking for microbiology culture
	2	Antigen presenting cells and phagocytosis	Microscope
	3	General bacteriology 1	Culture media 1
	4	General bacteriology 2	Culture media 2
	5	Microbial genetics I	Staining
	6	Microbial genetics II	
	7	General virology	
	8	General Mycology	
	9	Antimicrobial resistance	
HIS Module	1	Antigens and immunoglobulins	Serology 1
	2	Adaptive immunity and MHC	
	3	Complement system and cytokines	Serology 2
	4	Hypersensitivity	
	5	Tolerance, autoimmunity & tumor escape Mechanism	
	6	Lecture of transplantation and immunodeficiency	
	7	Overview of vaccinations	
MSS module	9	Clostridium & anaerobiosis	Methods of anaerobiosis

Parasitology

Description:

Parasitology is the study of parasites, their hosts, and the relationship between them. Parasitology will be learned as lectures and practical small-group sessions throughout 8 weeks. Continuous exam will be in the form of MCQ, and final exam will include practical exam and written exam in the form of MCQ and short answer questions.

Topics:

	Week No	Lectures	Practical
PPPM Module	1	General concepts of parasitology	Urine analysis for parasites
	2	Helminthology+ protozoology	
	3	Entomology+ myasis	Stool analysis for parasites
HIS Module	4	Filariasis	
	5	Leishmaniasis	Filariasis & Leishmania
	6	Malaria	Malaria
MSS module	7	Trichinosis	Trichinosis
	8	Cysticercosis	Cysticercosis

Early clinical experience 1

From your First day as a student you have professional obligations placed upon you by the public, the law and your colleagues which continue throughout your working life. Patients want more than merely intellectual and technical proficiency. The skills of history taking and physical examination are central to the practice of clinical medicine. The purpose of this early clinical experience is to explain how to:

- Take the history from a patient
- Examine a patient (general examination with stress on vital signs, chest, cardiac and abdominal examination)

	Week No	Lectures	Practical
ECE Module	1	History	
	2	Symptomatology	
	3	General examination part I	
	4	General examination part II	
	5	Vital signs I	
	6	Vital signs II	
	7	Abdominal examination	
	8	Cardiac examination	
	9	Chest examination	

Incision academy

Incision Academy is a surgical education platform. It provides access to a wide range of courses and procedures, filmed in 2D and 3D with over 500 video courses, interactive 3D anatomy models, and step-by-step learning modules. The procedures on incision platform are accredited by the Royal College of Surgeons of England. The chosen courses of the semester program are tailored specifically to be in accordance to the semester curriculum. Student will select 10 courses from the list.

Courses:

1. Preparation for Physical Examination
2. General Examination and Vital Signs
3. Examination of the Cardiovascular System
4. Examination of the Respiratory System
5. Surgical Anatomy of the Carpal Tunnel
6. Surgical Anatomy of the Wrist
7. Surgical Anatomy of the Hip
8. Surgical Anatomy of the Knee
9. Arterial Examination of the Upper Limb
10. Arterial Examination of the Lower Limb

Electronic portfolio

Medical student's E-Portfolio represents a collection of evidence of what you have been learned throughout medical school years both professionally and personally plus the student reflections on the education process.

Tasks:

1. Upload your semester 1 result and write a reflective piece about it.
2. Upload the results of 2 quizzes in semester 2.
3. Write a reflective piece on TBL session.
4. Upload a picture of your experiment on virtual physiology software.

الجدول التالي يوضح تقديرات المقررات

Marking system	Approximate Equivalent in ECTS	Definition According to ECTS Grading system
85 - 100 %	A	Excellent
75 to < 85%	B	Very good
65 to < 75%	C	Good
60 to < 65%	D	Satisfactory
30 to < 60%	F	Insufficient

Fundamental physical examination skills

ECE

Chest examination

Heart examination

Abdominal examination

Microbiology

Obtaining and handling a blood sample for culture (PPPM)

Taking swabs for different diagnostic purposes (PPPM)

Identifying different bacteria and fungi under the microscope (PPPM, MSS)

Differentiating different bacterial growth in culture (PPPM, MSS)

Anatomy

Dissecting the different parts and organs of the human body (MSS)

Parasitology

Identification of parasites and parasitic ova under the microscope (PPPM, HIS, MSS)

Preparing urine and stool specimen for microscopic examination (PPPM)

Histology

Identification of different normal tissue sections under the microscope (HIS, MSS)

Preparing and examining blood films and assessing hemoglobin value in a blood sample (HIS)

Pathology:

Identification of different pathological alterations in tissue sections under the microscope (PPPM, HIS, MSS)

Identification of gross pathological alterations in different body organ specimens (PPPM, HIS, MSS)

Physiology:

Determining blood group and performing cross matching and computability tests (HIS)

Biochemistry:

Practical laboratory techniques (Pipetting; basic laboratory handling skill).

Checklist FOR Obtaining and handling a blood sample for culture

Structure skill description	Process The action that the candidate demonstrates competency during the skill practice	Outcome Desired skill outcome
Preparation	▪ Verify patient identity and explain procedure. ▪ Perform hand hygiene and wear gloves. ▪ Assemble equipment: blood culture bottles (aerobic & anaerobic), sterile needles/syringe or vacutainer, alcohol swabs, chlorhexidine/povidone-iodine, tourniquet, sterile gauze, sharps container. ▪ Label bottles with patient name, ID, date, and time.	Proper preparation and labeling of culture bottles
Site Selection	▪ Select appropriate vein (e.g., median cubital). ▪ Apply tourniquet above venipuncture site. ▪ Cleanse site: first with 70% alcohol, then with chlorhexidine/povidone-iodine in concentric circles; allow to dry completely.	Sterile venipuncture site prepared
Venipuncture & Collection	▪ Perform venipuncture using aseptic technique. Collect recommended volume: 8–10 mL per bottle (adult), 1–3 mL (pediatric). ▪ Inoculate blood directly into culture bottles (aerobic first if using syringe; order doesn't matter if using vacutainer). ▪ Do not change needles between vein puncture and bottle inoculation.	Correct volume of blood collected aseptically into culture bottles
Post-Collection Handling	▪ Gently invert bottles to mix blood with broth. ▪ Do not refrigerate; keep at room temperature (20–25°C). ▪ Transport to lab within 2 hours (max 8 hours if delay unavoidable). Dispose of sharps in safety container. ▪ Remove gloves and perform hand hygiene.	Safe handling and timely transport of uncontaminated blood culture sample
Documentation	▪ Record patient details, date/time of collection, site of collection, and collector's name. ▪ Note any antimicrobial therapy patient is receiving	Complete and accurate documentation for laboratory and clinical correlation

Checklist FOR Identifying different bacteria and fungi under the microscope

Structure skill description	Process The action that the candidate demonstrates competency during the skill practice	Outcome Desired skill outcome
Preparation	- Place a drop of immersion oil directly on the prepared smear. - Rotate the nosepiece until the oil immersion objective lens (100x) snaps into position.	Proper preparation of slide and microscope for oil immersion.
Adjustment of Microscope	- Look through ocular lenses and lower the oil immersion lens carefully until it nearly touches the slide. - While viewing through the eyepiece, use only the fine adjustment knob to focus by moving lens up and down in small increments	Clear visualization of microorganisms under the microscope.
Assessment of Bacteria	- Observe stained bacterial smear. - Identify Gram reaction (positive/negative). - Determine shape (cocci, bacilli, etc.). - Note arrangement (clusters, chains, pairs, singles). Example: <i>E. coli</i>: Gram-negative bacillus in singles/chains; <i>S. aureus</i>: Gram-positive cocci in grape-like clusters.	Correct identification and classification of bacteria.
Assessment of Fungi	- Examine fungal smear under 10x and 40x before oil immersion. - Identify morphological structures such as yeast cells, budding forms, hyphae, or spores. - Compare findings with known fungal morphology (e.g., <i>Candida albicans</i> shows budding yeast with pseudohyphae).	Recognition and differentiation of fungal elements.
Documentation	- Record observations: Gram reaction, morphology, arrangement (for bacteria), and fungal structures. - Report results clearly in lab record.	Accurate documentation for diagnosis and correlation.

Checklist FOR Differentiating different bacterial growth in culture

Structure skill description	Process The action that the candidate demonstrates competency during the skill practice	Outcome Desired skill outcome
Colony Morphology on Solid Media (e.g., Agar Plates)	- Observe colony shape: circular, irregular, filamentous, rhizoid. - Examine margin: entire (smooth), undulate (wavy). - Note elevation: flat, raised, convex, umbilicate. - Assess color/pigmentation: e.g., <i>Pseudomonas aeruginosa</i> (green), <i>Staph aureus</i> (yellow). - Record size, surface texture (smooth, rough, wrinkled, mucoid), and opacity (transparent, translucent, opaque).	Ability to describe and differentiate bacterial colonies based on morphology.
Growth Patterns in Liquid Media (e.g., Broth)	- Observe turbidity: uniform cloudiness = even growth. - Look for pellicle: thin film on surface (e.g., <i>Bacillus</i> species). - Note sediment: growth at bottom. - Check for flocculent growth: clumps or flakes suspended in broth.	Recognition of different bacterial growth patterns in broth culture
Hemolysis on Blood Agar	- Identify Alpha-hemolysis (α): partial lysis, greenish discoloration (e.g., <i>Streptococcus pneumoniae</i>). - Identify Beta-hemolysis (β): complete lysis, clear zone around colony (e.g., <i>Streptococcus pyogenes</i>). - Identify Gamma-hemolysis (γ): no hemolysis (e.g., <i>Enterococcus faecalis</i>).	Differentiation of bacteria based on hemolytic activity.
Growth on Indicator/Differential Media	- MacConkey agar (MAC): <i>E. coli</i> (lactose fermenter) → pink colonies - Non-lactose fermenters (e.g., <i>Salmonella</i>, <i>Pseudomonas</i>) → Pale colonies. - CLED agar: <i>E. coli</i> (LF) → yellow colonies. Non-lactose fermenters → pale colonies. - Other indicator media: XLD: <i>Salmonella</i> → red colonies with black center.	Correct differentiation of lactose fermenters (LF) and non-lactose fermenters (NLF) on indicator/differential media.

Checklist FOR Taking swabs for different diagnostic purposes

Structure skill description	Process The action that the candidate demonstrates competency during the skill practice	Outcome Desired skill outcome
Nasopharyngeal (NP) Swab	▪ Tilt patient's head back \~70°. ▪ Use flexible mini-tip swab. ▪ Insert parallel to palate until resistance (nostril–ear depth). ▪ Roll gently, leave a few seconds. ▪ Withdraw while rotating. ▪ Place into transport tube.	Adequate NP specimen collected.
Oropharyngeal (OP) Swab	▪ Instruct patient: do not eat, drink, smoke, or brush teeth for at least 30 minutes before sample. ▪ Ask patient to open mouth and say “ahh”. ▪ Insert swab to posterior pharynx & tonsillar pillars. ▪ Rub swab over both tonsillar areas and posterior pharyngeal wall. ▪ Avoid contact with tongue, teeth, gums. ▪ Place swab into transport tube.	Proper OP sample collected without contamination.
Rectal Swab	▪ Wear gloves, explain procedure. ▪ Insert sterile swab \~2–3 cm into rectum. ▪ Rotate gently to collect fecal material. ▪ Place in transport medium.	Reliable rectal sample.
Wound Swab	▪ Clean wound surface with sterile saline/gauze. ▪ Insert swab into wound base/edge. ▪ Rotate to absorb pus/exudate. ▪ Place in transport medium (aerobic ± anaerobic).	Representative wound swab
Hand Swab (Hygiene & infection control)	▪ Moisten sterile swab with sterile saline/transport medium. ▪ Rub swab firmly over palm, fingertips, and interdigital spaces. ▪ Rotate swab while covering multiple areas. ▪ Place into transport tube.	Hand flora sample collected.

Checklist for microscopic identification of parasites and parasitic ova under microscope

Steps		Yes	No
1	Put the slide correctly on the microscope stage.		
2	Turn on the light of the microscope.		
3	Begin examination by using the low-power objective lens.		
4	Use course and fine adjustment to clearly see the specimen.		
5	Use the high-power objective lens to identify details.		
6	Examine the whole slide in both directions (longitudinal and transverse).		
7	Accurately identify parasite or ova.		
8	Record findings.		
Total marks			

Checklist for stool sample microscopic examination

Steps		Yes	No
1	Wear personal protective equipment (coat, mask and gloves).		
2	Macroscopic examination of specimen (color, consistency, blood, mucous parts, and parasites or segments).		
3	Use a clean slide.		
4	Place a drop of saline on the center of the slide.		
5	Take a small amount of stool (a size of a matchstick head) with a wooden applicator stick.		
6	Mix this stool amount with saline by the applicator stick.		
7	Put a clean coverslip on the sample beginning with a 45° angle to spread it.		
8	Put the slide correctly on the light microscope and use it properly (examine the whole slide).		
9	Safe disposal of the slide and gloves after examination.		
10	Use alcohol to disinfect examination surface, microscope and hands.		
Total marks			

Checklist for urine sample microscopic examination

Steps		Yes	No
1	Wear personal protective equipment (coat, mask and gloves).		
2	Comment on physical appearance of urine sample (color, turbidity, and odor).		
3	Use plastic pipette (dropper) in sampling.		
4	Take the sample from the bottom of the conical tube.		
5	Use a clean slide.		
6	Put a drop of the sample on the center of the slide.		
7	Put a clean coverslip on the urine drop beginning with a 45° angle to spread it.		
8	Use the light microscope properly (examine the whole slide).		
9	Safe disposal of the slide and gloves after examination.		
10	Use alcohol to disinfect examination surface, microscope and hands.		
Total marks			

Pipetting skill assessment checklist

Questions	Student Mark	Marks
Select the appropriate pipette and tip for the required volume and firmly attach the tip.	0.5	
Hold the pipette vertically when aspirating and press the operating button down to the first stop.	0.5	
Dip the tip into the liquid to a depth of 1cm, slowly release the operating button and then withdraw the pipette from the liquid.	0.5	
Dispense the liquid into the receiving vessel by gently pressing the operating button down to the stop to empty the tip, remove the pipette from the vessel, and then waste the tip.	0.5	
Total	/2	

Blood group cross matching checklist

Wear White Coat & Gloves.	1
Identifies required materials correctly (anti-A, anti-B, anti-D sera, slides/test tubes)	1
Performs ABO blood grouping using correct technique and labeling	2
Correctly observes and interprets agglutination for ABO grouping	1
Performs Rh (D) typing correctly and safely	1
Correctly interprets and records blood group (ABO and Rh)	1

Examination of a blood Smear (Checklist)

A. Quality of the blood smear		
A: Determine the quality of the blood smear? Good bl. Film or bad bl. film	Answer:	(1 mark)
B: Give 2 reasons? <u>Good BL. Smear:</u> occupy central part of slide, regular, thin and not cover entire slide <u>Bad BL. Smear:</u> doesn't occupy central part of slide, irregular, thick or cover entire slide	Answer: 1: 2:	(2 marks)
B. Examination of blood smear		
Mention the stain of the blood smear? Leishman stain	Answer:	(1 mark)
Examine the stained blood smear with the oil immersion lens and identify the blood leucocyte? Neutrophil, Lymphocyte, eosinophils or basophil	Answer:	(1 mark)
Total Mark		/ 5

قواعد الانتقال بين الفصول والمستويات والمراحل

- يعد الطالب ناجحاً في المقرر الدراسي إذا حصل على 60% من الدرجات الكلية للمقرر و 40% في درجات الامتحان النظري النهائي.
- المقررات الاختيارية: إذا حصل على 50% من الدرجات الكلية للمقرر يعتبر ناجحاً ولا تضاف الدرجات في المجموع.
- يسمح للطالب الراسب في الدور الأول بدخول امتحان المقرر.
- في حالة رسوب الطالب في مقررات بالمرحلة الأولى أو بالمرحلة الثانية فإنه ينتقل إلى المستوى الأعلى إذا لم تزد عدد النقاط التي رسب بها عن 22 نقطة معتمدة.
- في حالة تعدي عدد النقاط المعتمدة لا ينتقل الطالب إلى السنة التالية ويبقى لإعادة بنفس السنة لحين استيفاء النقاط المقررة.
- في حالة عدم اجتياز الطالب مقرر أو أكثر بالمرحلة الأولى (Semester 1 to 5) يسمح له بالتسجيل لمقررات الفصل الدراسي السادس على أن يعيد امتحانات المقررات التي رسب بها في الدور الثاني
- في حالة عدم اجتياز هذه المقررات لا يسمح للطالب بالتسجيل في مقررات المرحلة الثانية إلا بعد الانتهاء من جميع مقررات المرحلة الأولى.
- الطالب الراسب يحسب له حد أقصى 64.9% من الدرجة الكلية للمقرر في امتحان الدور الثاني وعند إعادة دراسة المقرر.
- طلاب المستوى الثاني الحد الأقصى للدراسة ثلاث سنوات في المستوى الواحد سنة دراسية كمستجد وستين آخرين كباقي لإعادة.

GOOD LUCK

